VJC 2022 H2 Biology Prelim Paper 2 Answer

Question 1

(a) (i) Identify organelles **A** and **B** and explain their roles in beta cells. [3]

Identifying organelles
- A – nucleolus, B – mitochondrion;

Role of nucleolus
- Produce ribosomes for protein synthesis of insulin;

Role of mitochondrion
- Produce ATP for synthesis and exocytosis of insulin;

(ii) With reference to Fig. 1.1 and 1.2, explain how the cell theory is supported. [2]

- Islet of Langerhans consists of many cells – living organelles consist of one or more cells;
- Cell contains organelles for ATP production, DNA for inheritance, etc. – cell is the most basic unit of life;

(iii) Outline the roles of the intermembrane system in the synthesis of insulin within beta cells. [3]

- Translation of insulin polypeptide at ribosomes on rough endoplasmic reticulum;
- Folding of insulin protein in cisternae of rough endoplasmic reticulum;
- Budding of transport vesicle to transport insulin from rough endoplasmic reticulum to Golgi apparatus;
- Biochemical modification of insulin in Golgi apparatus;

(iv) Describe how insulin is released by beta cells. [2]

- Insulin released via exocytosis;
- Insulin granules move to and fuse with cell surface membrane;

(b) With reference to Fig. 1.3, explain how the structure of insulin is significant to its role in the body. [3]

- Globular;
- Soluble in the blood to be transported to target cells;
- Specific 3D shape complementary to shape of insulin receptor;

Question 2

(a) Using a labelled diagram, show how a triglyceride molecule is hydrolysed. [3]

- Correct drawing and labels of triglyceride, glycerol and fatty acids;
- Correct labelling ester bonds and indication of hydrolysis sites;
- Correct label of hydrolysis and showing that 3 molecules of water are used;

(b) (i) Describe the structure of the fatty acid transporter and explain how it supports its role. [3]

- Ref. to amino acids with non-polar R groups forming hydrophobic interactions with hydrocarbon tails of phospholipids;
- Ref. to amino acids with polar R groups forming hydrogen bonds with hydrophilic phosphate heads and aqueous medium;
- Ref. to water-filled pore / channel for fatty acid to pass through;

(ii) Account for the number of ATP molecules produced for each acetyl CoA that enters the TCA (Krebs) cycle. [3]

- Ref. to 1 ATP, 3 NADH and 1 $FADH_2$ produced per acetyl CoA;
- Each NADH yields 2.5 ATP and each $FADH_2$ yields 1.5 ATP via the ETC;
- $1 + (3 \times 2.5) + (1 \times 1.5) = 10$ ATP produced;

(iii) Explain why, for the same mass of glucose and triglyceride respired, a lot more ATP is produced from the triglyceride. [2]

- Ref. to long hydrocarbon tails in triglycerides which yield many acetyl CoA in beta-oxidation;
- Ref. to each glucose yielding only 2 acetyl CoA;

Question 3
(a) Briefly describe the role of DNA polymerase III in a eukaryotic cell. [2]

- Addition of deoxyribonucleotides to 3’ OH via formation of phosphodiester bonds;
- Elongation from RNA primer to form daughter DNA strand;
- Ref. to proofreading;

(b) (i) With reference to their molecular structures, explain why the three polymerases have a different optimal pH. [2]

- They have different numbers of ionic bonds;
- Polymerases with more ionic bonds are more likely to denature as the pH changes;

OR

- They may have different R-groups;
- Polymerases with R-groups that can take up or release H^+ ions will be able to work well even under extreme pH;

(ii) Explain why P-Omega has a high percentage efficiency at 90 °C. [2]

- Has a higher proportion of disulfide linkages, does not denature at high temperatures;
- The higher kinetic energy at higher temperatures increases the rate of collision between enzyme and substrate and therefore increases the rate of reaction;
- Enzyme thus operates more efficiently;

(c) (i) Suggest which polymerase the researcher should choose to replicate his DNA. Explain the reasons behind your choice. [2]

- P-Beta;
- Works relatively efficiently in an alkaline medium;
- Shows some activity even at 90°C, thus it is not completely denatured at high temperature;

(ii) Fill up Table 3.2 below to indicate the types of polymerase from each of the sources. [1]

type of polymerase	**origin of polymerase**
P-Alpha	yeast
P-Beta	microbes from Bonneville Salt Flats
P-Omega	volcanic lake in New Zealand

Question 4

(a) Using your knowledge on DNA replication in cells, explain how the Hayflick limit occurs. [4]

- Ref. to end-replication problem;
- RNA primer removed at 5' end of daughter DNA strand;
- DNA polymerase cannot replace it with DNA nucleotides due to a lack of 3' OH;
- The telomeres shorten to the critical length that stops further cell division;

(b) (i) With reference to Fig. 4.1, describe how pHi becomes acidic. [3]

- Ref. to histone deacetylation by HDAC;
- Acetyl groups broken down into acetate and H^+;
- H^+ concentration increases in the cytoplasm thus decreasing pHi;

(ii) Suggest a role for such changes in histone acetylation other than transcriptional regulation. [1]

- Regulate / Maintain pHi;

(c) With reference to Fig 4.2, identify and describe the post-translational control of the amount of functional GPCRs. [4]

- Biochemical modification in the rough endoplasmic reticulum and Golgi apparatus;
- GPCR transported from rough endoplasmic reticulum to Golgi apparatus;
- Mature GPCR transported from Golgi apparatus to the cell surface membrane;
- Misfolded GPCR undergoes protein degradation by proteasome;
- Dimerisation of GPCR results in internalisation, placing the dimer into an endosome;
- Endosome (MVB) fuses with lysosome to digest the dimer;

Question 5

(a) Account for the difference between the cDNA obtained from the RT-PCR of a eukaryotic mRNA and the gene from which the mRNA was transcribed. [2]

- cDNA lack introns;
- cDNA is reverse transcribed from mature mRNA which had undergone splicing;

(b) (i) State which mutant has a phenotype consistent with a loss-of-function mutation in the *trpR* gene. Explain your answer. [3]

- Mutant 1;
- Loss-of-function mutation results in non-functional repressor produced;
- Unable to inhibit expression of *trp* operon regardless of the presence of tryptophan;

(ii) State the component of the *trp* operon in mutant 3 in which a loss-of-function mutation has occurred. Explain your answer. [2]

- Promoter;
- RNA polymerase cannot bind to initiate transcription;

(iii) If the phenotype of mutant 3 is caused by a mutation in the *trpR* gene, explain how this mutation would affect the structure and function of the repressor protein. [3]

- Ref. to a missense mutation;
- Change in 3D conformation of repressor produced, which is in the active conformation
- Able to bind to operator to inhibit expression of *trp* operon regardless of the presence of tryptophan

Question 6

(a) The M checkpoint occurs during mitosis of the cell cycle. Explain the significance of this checkpoint. [2]

- Checks for proper binding of spindle fibres to centromeres of chromosomes;
- Ensure proper separation of chromosomes / sister chromatids to prevent non-disjunction;

(b) With reference to Table 6.1, account for the differences in percentage distribution of cells at each stage. [2]

- Majority of the cells (87.3%) are in interphase stage;
- Interphase is the longest stage due to G1 checkpoint checking for presence of appropriate growth signals / time taken to carry out DNA replication;

(c) Using information from Table 6.1 and 6.2, explain the effect of oncogenes on cell division. [2]

- Fewer cells remain in interphase (42.9%) / More cells are in mitosis (57.1%) in the tumour;
- Oncogenes result in over-stimulation of cell division;

(d) Suggest why it is important to use percentage values when comparing data from the normal and tumour cell cultures. [1]

- The initial number of cells in each culture may not be the same;

Question 7

(a) (i) State the genotypes of the F1 male moth and the F2 female moth with white body and long antenna. [1]

- F1 male moth – $Z^{A}_{B}Z^{a}_{b}$, F2 female moth with white body and long antenna – $Z^{a}_{B}O$;

(ii) Explain how the F2 female moth with white body and long antenna is obtained. [3]

- Crossing over occurs in prophase I in the F1 male moth;
- Producing a gamete with the Z^{a}_{B} chromosome;
- Fertilisation of this gamete with a gamete with no Z chromosome from the female;

(b) In the space provided below, draw a genetic diagram of the above cross to show the expected phenotypic ratio of the offspring. [4]

You do not have to include gene **B** or antenna length in the diagram.

F1	phenotype	male with yellow body			x	female with white body				
	genotype	Z^AZ^add				Z^aODD				1m
					meiosis					
	gametes	Z^Ad		Z^ad		Z^aD		OD		1m
F2	genotype	Z^AZ^aDd		Z^AODd		Z^aZ^aDd		Z^aODd		1m
	phenotype	male with brown body		female with brown body		male with white body		female with white body		
	phenotypic ratio	1	:	1	:	1	:	1		1m
	observed numbers	48	:	51	:	50	:	51		

Question 8
(a) With reference to Fig. 8.2, comment whether

(i) beak depth is an inherited characteristic. [2]

- Yes – as mean beak depth of parents increased from 7 to 11 mm, mean beak depth of offspring increased from 7.5 to 10 mm / mean beak depth of offspring are about the same as their parents;
- Parents with alleles coding for higher beak depths passed their alleles to their offspring;

(ii) more than one gene controls beak depth. [1]

- Yes – there is continuous variation in beak depth;

(b) (i) Explain why the range of beak depth of the medium ground finch on Isabela Island is different from those on Daphne Island. [5]

- Variation of beak depth within the populations on both islands exists due to mutations;
- Presence of smaller-sized seeds on Daphne Island;
- Those with smaller-sized beaks were selected for on Daphne Island as they were better able to feed on smaller-sized seeds;
- Those with selective advantage are more likely to survive, reproduce and pass on the alleles coding for beak size to their offspring;
- Frequency of alleles coding for smaller-sized beaks in the Daphne Island population increased over time and more individuals have smaller-sized beaks;

(ii) Outline how the medium ground finch on Daphne Island may evolve as a distinct species thousands of years in the future. [3]

- No gene flow between the two populations due to long distance between them;
- Natural selection continues within each population and mutations arise independently, such that over time, the two populations accumulate changes;
- They are classified as different species as they can no longer interbreed to form viable offspring;

(iii) Suggest why both species of finches are found on Isabela Island while only one is found on the other islands. [1]

Any one
- Medium ground finch may have originated on Daphne Island, small ground finch on Crossman Island and some members of both species migrated to Isabela Island;
- Both originated on Isabela and only one species happened to migrate to one of the other islands;
- Isabela Island is much bigger than the other two islands and likely have a variety of different-sized seeds to support both species;

Question 9
(a) Contrast between an antigen and an antibody. [2]

features	antigen	antibody
origin	found on / in a pathogen	produced by a plasma cell / found on cell surface membrane of B cell
composition	may be protein, carbohydrate, lipid in nature	protein
binding	has an epitope that binds to antigen binding site on antibody	contains antigen binding site for binding of antigen

(b) (i) Explain how the structure of the cell surface membrane of dendritic cell allows its role as an antigen presenting cell. [3]

- Ref. to the presence of receptors recognising a broad class of antigens;
- Ref. to weak hydrophobic interactions between fatty acid tails of phospholipids giving rise to the fluidity of cell surface membrane for phagocytosis;
- Ref. to the presence of MHC molecules for antigen-presentation;

(ii) Outline how immunological memory is developed after activation of specific helper T cells. [4]

- Activated helper T cells release cytokines;
- Specific B cell activated by cytokines to differentiate into memory B cells;
- Ref. to memory B cells recognising the same antigens found on the actual pathogens;
- Infection by the same pathogen activates the memory B cells to differentiate into plasma cells;
- Plasma cells release antibodies against the pathogen;

Question 10

(a) With reference to Fig 10.1 and 10.2, other than missing ice caps, predict how the natural landscape of the Swiss Alp would change in the next decade. [3]

- Ref. to increasing trend of temperature – temperature expected to continue increasing;

Any two

- New mountain lakes / rivers will appear as the ice melts away;
- Melting glaciers will contribute to sea level rising;
- New plants with higher heat tolerance will grow on the mountain tops;
- Presence of new habitats and warmer coonditios will encourage migration of animals uphill;
- Displacement of species living at the peak can lead to extinction;

(b) (i) State the optimal temperature for growth and development of the stone pine seedlings [1]

- 12 (accept 11-13) ºC;

(ii) With reference to Fig 10.3, describe the change in rates between the gross and net photosynthesis before 35ºC. [2]

- Both gross and net photosynthesis rates increase from -5 to 12ºC;
- Gross photosynthesis rate continues to increase and plateaus around 35ºC, net photosynthesis rate drops till it reaches zero at 35ºC;

(iii) With your knowledge on photosynthesis, explain the change observed in **(bii)**. [3]

- From -5 to 12ºC, increasing temperatures increased kinetic energy of enzymes in Calvin cycle, increasing the formation of enzyme-substrate complexes, thus increasing the gross and net photosynthesis rates;
- Beyond the optimal temperature of around 12ºC, denaturation of the enzymes diminishes the gross photosynthesis rates;
- And net photosynthesis rate decreased rapidly as respiration rate increased;